INCH-POUND
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June 28, 2006

COMMERCIAL ITEM DESCRIPTION

T-SHIRT, MOISTURE-WICKING, ARMY

The General Services Administration has authorized the use of this commercial item description for use by all federal agencies.

1. SCOPE.

This Commercial Item Description covers the requirements for a moisture-wicking T-shirt.

2. CLASSIFICATION.

The T-shirt shall be available in the following classes and sizes as specified:

Class 1 - Desert Sand 503
Class 2 - Tan 499

2.1 Sizes

XX-Small (XXS)	Large (L)
X-Small (XS)	X-Large (XL)
Small (S)	XX-Large (XXL)
Medium (M)	XXX Large (XXXL)

3. SALIENT CHARACTERISTICS.

3.1 Description. The T-shirt shall have short sleeves, a crew neck collarette that shall be 1 ± 1/8 inch wide and hemmed sleeves and bottom. The sleeve shall be hemmed 3/4 ± 1/8 inch wide and the bottom hem shall be 7/8 ± 1/8 inch wide. All seams shall be overedge or cover stitched. The inside of the shoulder and back of collar seams (shoulder to shoulder) shall be reinforced with binding tape 3/8 to 1/2 inch wide. An identification/size/care label shall be caught in the center back collar seam.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: DLA Troop Support, Clothing and Textiles Directorate, Attn: FQSE, 700 Robbins Ave, Philadelphia, PA 19111. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8415

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3.2 Visual Shade Matching. The color and appearance of the material shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 ± 200 K illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 ± 200K.

3.3 Materials.

3.3.1 Basic material. The basic material shall be 100% polyester tubular jersey knit constructed with a staple yarn for the body and sleeves. The cloth shall have moisture management capability and shall conform to the requirements in Table I.

3.3.2 Collarette. The material for the collarette shall be a l x l true rib knit with a weight of 7.2 +/- 0.2. and shall meet or exceed the physical properties of the basic material as specified in Table I.

TABLE I
Basic Material requirements

CHARACTERISTIC	REQUIREMENT	TEST METHOD
Fiber Identification	100% Polyester	ASTM D 276 or AATCC 20
Weight, oz/sq yd.	5.3 ± 0.2 basic 7.2 ± 0.2 rib	ASTM D 3776, Opt C
Bursting strength lbs (min.)	120	ASTM D 3787
Dimensional stability percent (max.) Home laundering: 5 cycles Wales Courses	 4.0 4.0	 AATCC 135, Table I (3,V, Aiii)
Colorfastness rating (min.) Laundering Light (40 hours or 170 kJ))	 3-4 3-4	 AATCC 61 1A 1/2/ AATCC-16, Opt 1 or 3 1/
Moisture vapor transmission rate g/m^2/24 hours (min) Initial After 5 Launderings	 1550 1450	ASTM E 96, Test "B" 3/ AATCC 135 Table I (3,V,Aiii) 4/ & ASTM E 96, Test "B" 3/
Seam smoothness (min.)	SS-3	AATCC 88B Table II (3,V,Aiii)
Seam strength lbs (min.)	70	ASTM D 1683 5/
Pilling (min.)	3	ASTM D 3512 6/
Air permeability, ft^3/ min./ft^2 (min.)	280 basic material 250 rib	ASTM D 737
Wicking, inches (min)	3 1/2	See 3.5
Toxicity	7/	See 3.6

1/ AATCC Evaluation Procedure 1, Gray Scale for Color Change

2/ AATCC Evaluation Procedure 2, Gray Scale for Staining

3/ The back side of the cloth shall face the water; the free stream air velocity shall be 550 ± 50 FPM as measured 2 inches above the fabric specimen. The air flow shall be measured at least 2 inches

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from any other surface. The test shall be run for 24 hours and weight measurements shall be taken at only the start and completion of the test. At the start of the 24 hour test period, the air gap between the water surface and the back of the specimen shall be $^{3}/_{4} + {}^{1}/_{16}$ inch.
4/ The last wash cycle shall be performed without the use of detergent. The specimens shall be dried after each laundering.
5/ The procedure for ASTM D 1683 shall be followed except that knit seam strength shall be evaluated, the test shall use a 3 inch gauge length. Seam strength shall be measured at the shoulder and underarm seams.
6/ Pilling visual appearance shall be measured using the scale in ASTM D 3512
7/ The finished cloth shall not present a dermal health hazard when used as intended

3.3.3 Thread. Thread for all seaming and stitching shall be in accordance with standard commercial practice. The color of the thread shall be a good match to the basic material. The finished thread shall have no chemical finishes or treatments, other than those commonly used on commercial threads.

3.4 Labels

3.4.1 Label/marking. A commercial type label shall include size, contractor's name, contract number, National Stock Number (NSN), fiber and laundering (care) information. The label shall have rating of 3-4 for colorfastness to laundering when tested in accordance with AATCC-61, Test 3A.

3.4.2 Bar code label. Each item shall be individually bar-coded with a paper tag for personal clothing items. The paper tag shall be standard bleached sulfate having a basis weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each item by a fastener, or the barcode and Universal Product Code (UPC) tags may be attached as a sticker to the outside of the individual poly bag. The tags shall be clearly legible and readable by a scanner. The bar coding element shall be a 13 digit national stock number (NSN). There shall be a 12 digit UPC number assigned for all NSN's by the contracting activity. The initials "UPC" must appear beneath the code. The bar code for NSN and UPC type shall be a medium to high density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and so it causes no damage to the item. This UPC code must also be placed on all shipping cartons on which the NSN appears.

3.5 Wicking Test Procedure. The wicking test procedure is used to determine the rate at which water will wick on test specimens suspended in water.

3.5.1 Apparatus

3.5.1.1 Flask. A 500 ml Erlenmeyer Flask shall be used.

3.5.1.2 Straight Pins. Straight pins approximately 3 inch length shall be used.

3.5.1.3 Food Coloring. A food color shall be used that will be visible on the specimen to indicate the water level during testing.

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3.5.2 Test procedure. Fill a 500 ml ERLENMEYER flask with 200 ml colored water (fill as many flasks as specimens to be tested.) Cut 6 x 1 inch strip of specimens to be tested (6 inch length is cut in the wales direction). Pierce top edge of strip (approximately $^{1}/_{8}$ inch to $^{1}/_{4}$ inch from top) with long straight pin as shown: Suspend strip from pin in flask filled with 200 ml colored water, as shown in Figure 1. Remove the strip from the flask after one hour has elapsed. Measure water level on strip in inches and record.

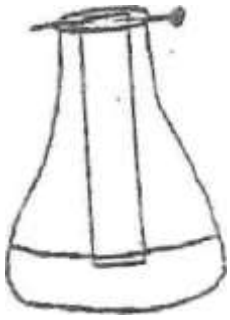


FIGURE I

3.6 Toxicity test. The contractor must furnish information, which certifies that the finished product is composed of materials, which have been safely used commercially or provided sufficient toxicity data to show compatibility with prolonged, direct skin contact. At a minimum, toxicity data should include results from a primary dermal irritation study in laboratory animals and a repeated insult human patch test (Modified Draize Procedure). The latter must be conducted under the supervision of a qualified dermatologist using at least 100 free-living individuals.

3.6.1 Toxicity documents. All finishes/chemicals used to process the T-shirt shall be identified and accompanied by the appropriate Material Safety Data Sheet (MSDS) information. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.7 Figure. Figure 2 is furnished for informational purposes only. To the extent of any inconsistencies between the written document and the figure, the written document shall govern.

3.8 Finished measurements. The T-shirt shall conform to the finished measurements specified in Table II.

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TABLE II
Finished measurements (inches)

SIZES	1/2 Chest (A)	Back Length (B)	1/2 Neck Opening (C)	Shoulder (D)	Armhole Opening (E)	Under Sleeve (F)	Over Sleeve (G)	Sleeve Opening (H)
XX-Small	14	25	6-1/2	2	9-1/2	2-3/4	5-3/4	6-1/2
X-small	16	26	7	3	9-1/2	2-3/4	5-3/4	6-1/2
Small	18	27	7-1/2	4	9-1/2	2-3/4	5-3/4	6-1/2
Medium	20	28	8	5	10	3-1/2	6-1/2	7
Large	22	29	8-1/2	6	10-1/2	4-1/4	7-1/4	7-1/2
X-Large	24	30	9	7	11	5	8	8
XX-Large	26	31	9-1/2	8	11-1/2	5-3/4	8-3/4	8-1/2
XXX-Large	28	32	10	9	12	6-1/2	9-1/2	9
Tolerance	± 1/2	+1, -1/2	± 1/4	± 1/4	± 1/4	± 3/8	±1/2	± 1/4

Method of measuring:

(A) 1/2 Chest: Lay T-shirt flat but not stretched, measure across from folded edge to folded edge, approximately 1 inch below the armhole (see Figure 2).

(B) Back Length: Lay T-shirt flat but not stretched, measure from collar joining seam on center back to bottom of shirt (see Figure 2)

(C) Neck Opening: Grasp collar at center back and center front, forming a straight line, but do not stretch, and measure.

(D) Shoulder: Measure from collar seam to armhole seam, along shoulder seam (see Figure 2).

(E) Armhole Opening: Lay T-shirt flat but not stretched, and measure from the intersection of top of sleeve and shoulder seam to bottom of underarm seam.

(F) Under Sleeve: Lay T-shirt flat but not stretched, measure from bottom of underarm seam to end of sleeve along seam.

(G) Over Sleeve: Lay T-shirt flat but not stretched, measure from the top of sleeve at shoulder seam to the bottom of sleeve.

(H) Sleeve Opening: Lay T-shirt flat but not stretched, measure from folded edge to folded edge along bottom of sleeve.

3.9 Patterns. Commercial patterns shall be used and shall conform to the design requirements cited in this document.

4. REGULATORY REQUIREMENTS.

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description and conform to the producer's own standards and quality assurance

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practices used for similar products offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Visual examination. Each T-shirt shall be examined for the following defects:

Material defects Any hole, cut, tear, mend, drop stitch, loose knit, miss knit, slub, streak, knitted in waste, permanent fold, pleat or crease; any spot or stain, discoloration or color not as specified, any shaded part, any objectionable odor
Seams and stitching Loose or tight stitching tension, broken or missing thread or stitch, needle chew; any part caught in an unrelated operation or stitching; raw edge, thread breaks, thread ends not removed. Any seam puckered, distorted, pleated, irregular or open.
Workmanship Any component part omitted, twisted, not specified size, missing, misplaced, poorly shaped or not laying flat. Measurement of item not as specified; sleeves differing in lengths by more than 3/8 inch; design not as specified.
Labels Label(s) missing, incorrect, or illegible; bar code or UPC code label omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded, packaged item; bar code label causes damage to the item.

5.3 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order.

6. PACKAGING.

Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of documents.

7.1.1 Copies of government documents are available online at https://assist.dla.mil.

7.1.2 Copies of documents are available online at www.astm.org or from the ASTM INTERNATIONAL, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.1.3 Copies of documents are online at www.aatcc.org or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.

7.1.4 Federal Acquisition Regulations are available online at http://acquisition.gov/far/index.html or by contacting the Superintendent of Documents at 202-512-1800.

7.1.5 MISCELLANEOUS Principle and Methods of Toxicology, A Wallace Hayes (editor), pp 394-396, 1989. (Copies of this document is available from Raven Press, 1185 Avenue of the Americas, New York, NY 10036)

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Marzulli, F. and H. Maibach, “Contact Allergy: Predictive Testing in Humans,” Advances in Modern Toxicology, Volume 4, pp 353-372, 1977. (Copies of this document are available from the U.S. Army Center for Health Promotion and Preventative Medicine, ATTN: MCHB-DC-TTE, Bldg., E-2100, Aberdeen Proving Ground, MD 21010-5422.)

7.2 For access to standard samples, address the contracting activity issuing the invitation for bids or request for proposal. Standard samples are also available at DLA Troop Support through http://warfighter.dla.mil under tab “Vendor Info” then “Specifications/Pattern Request” under “Special Instructions” provide color shade, roll number and solicitation/contract number)

7.3 Ordering Data. The contract or order should specify the following:

a. CID document number and revision.
b. Sizes required
c. Product conformance provisions.
d. Packaging requirements.

7.4 Key Words:

Polyester
Undergarment

MILITARY INTERESTS:

Custodians
Army - GL

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
DLA - CT
Project 8415-2020-015

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FIGURE II
T-Shirt, Moisture-Wicking, Army

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